

Introduction & Skeletal System

Core Learning · Foundations of Human Anatomy & Physiology

Section 1: Anatomical Terminology & Planes of Movement

A common anatomical language allows coaches to communicate precisely with other health professionals (physiotherapists, sports physicians, and strength and conditioning coaches). It also helps when analysing movement and communicating observations about technique. All anatomical descriptions are made with reference to the anatomical position.

1.1 The Anatomical Position

The anatomical position is the standard reference point for all anatomical descriptions. It is achieved by:

1. Standing upright with the head facing forward
2. Arms hanging by the sides
3. Palms of hands facing forward (anteriorly)
4. Feet slightly separated with toes pointing forward

Using the anatomical position ensures that all movement descriptions are consistent and universally understood — essential when collaborating with physiotherapists and medical professionals.

1.2 Anatomical Terms of Location

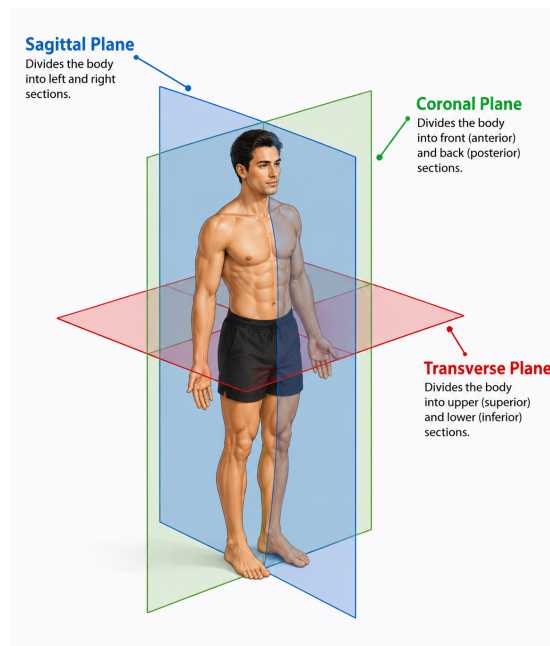
Term	Definition & Example
Anterior	The front — e.g. the chest is on the anterior surface of the body
Posterior	The back — e.g. the spine is on the posterior part of the trunk
Superior	Toward the head — e.g. the shoulder is superior to the hip
Inferior	Toward the feet — e.g. the knee is inferior to the hip
Lateral	Toward the side/outside — e.g. the little toe is on the lateral side of the foot
Medial	Toward the midline/inside — e.g. the big toe is medial to the little toe
Proximal	Nearer the trunk — e.g. the elbow is proximal to the wrist
Distal	Further from the trunk — e.g. the fingertips are distal to the wrist
Superficial	Nearer to the surface — e.g. the skin is more superficial than the muscle
Deep	Further from the surface — e.g. the bone is deep to the muscle
Prone	Face down — e.g. lying on the stomach
Supine	Face up — e.g. lying on the back

1.3 The Three Planes of Movement

Movements of the human body are described in terms of the plane in which they occur. There are three planes and climbing is a multi-planar activity — coaches must understand all three:

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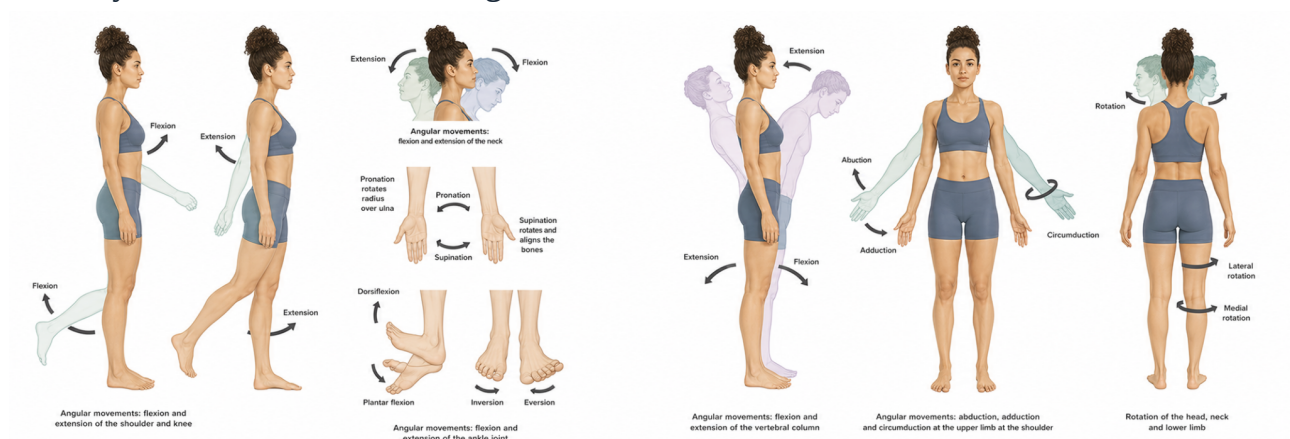


The sagittal, coronal, and transverse anatomical planes

Plane	Description	Climbing Examples
Sagittal plane	Passes from front to back, dividing the body into left and right halves. Front-to-back movements.	Pull-ups, squats, deadlifts, pushing movements, walking to the crag
Coronal plane (frontal)	Passes from side to side, dividing the body into anterior and posterior halves. Side-to-side movements.	Lateral step-through on a traverse, arm flagging to the side, lateral raises
Transverse plane	Passes horizontally, dividing the body into upper and lower halves. Rotational movements.	Drop-knee (hip rotation), cross through moves, trunk twisting on steep terrain

Most climbing movements are multi-planar. A single move on an overhang might simultaneously involve sagittal plane pulling, transverse plane hip rotation, and frontal plane flagging. This complexity is what makes climbing skill so challenging to learn and coach.

1.4 Key Joint Actions for Climbing Coaches



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Flexion, extension, rotation, abduction, adduction and circumduction

Joint Action	Definition	Climbing Application
Flexion	Decreasing the angle between two bones (bending)	Pulling into a lock-off — elbow and shoulder flexion
Extension	Increasing the angle between two bones (straightening)	Straightening the arm on a rest jug — elbow extension
Abduction	Moving a limb away from the midline of the body	Raising the leg out to the side for a high foothold
Adduction	Moving a limb toward the midline of the body	Drawing the leg back toward the body after a high step
Rotation	Turning a limb around its long axis (medial/lateral)	Hip rotation in a drop-knee; shoulder rotation on sidepulls
Circumduction	Circular movement combining flex, extension, abduct, adduct	Warming up the shoulder before a session
Pronation	Rotating the forearm so palm faces down/backward	Pulling on a jug with palm facing toward the climber
Supination	Rotating the forearm so palm faces up/forward	Undercling position — forearm supinated
Dorsiflexion	Moving top of foot toward the shin	Toe hook — ankle maximally dorsiflexed to lock onto hold
Plantar flexion	Moving top of foot away from the shin (pointing toes)	Toe down — getting maximal pressure onto a small hold
Protraction	Forward movement of the scapula (rounding shoulders)	Common postural fault in climbers from dominant pulling
Retraction	Backward movement of the scapula (squaring shoulders)	Scapular retraction exercises for antagonist training
Depression	Downward movement of the scapula	Initiating a pull-up correctly — scapular depression before elbow flexion
Elevation	Upward movement of the scapula (shrugging)	A common fault when fatigued — 'shrugging' into lock-off positions

Section 2: The Musculoskeletal System

The musculoskeletal system is the combination of the muscular and skeletal systems working together. It includes the bones, muscles, tendons, and ligaments of the body, and provides shape, protection of internal organs, and the ability to move. Every time you sit, stand, walk, or climb you are using the musculoskeletal system. When it is injured or not functioning properly, our ability to complete even everyday tasks is greatly reduced — something every injured climber knows firsthand.

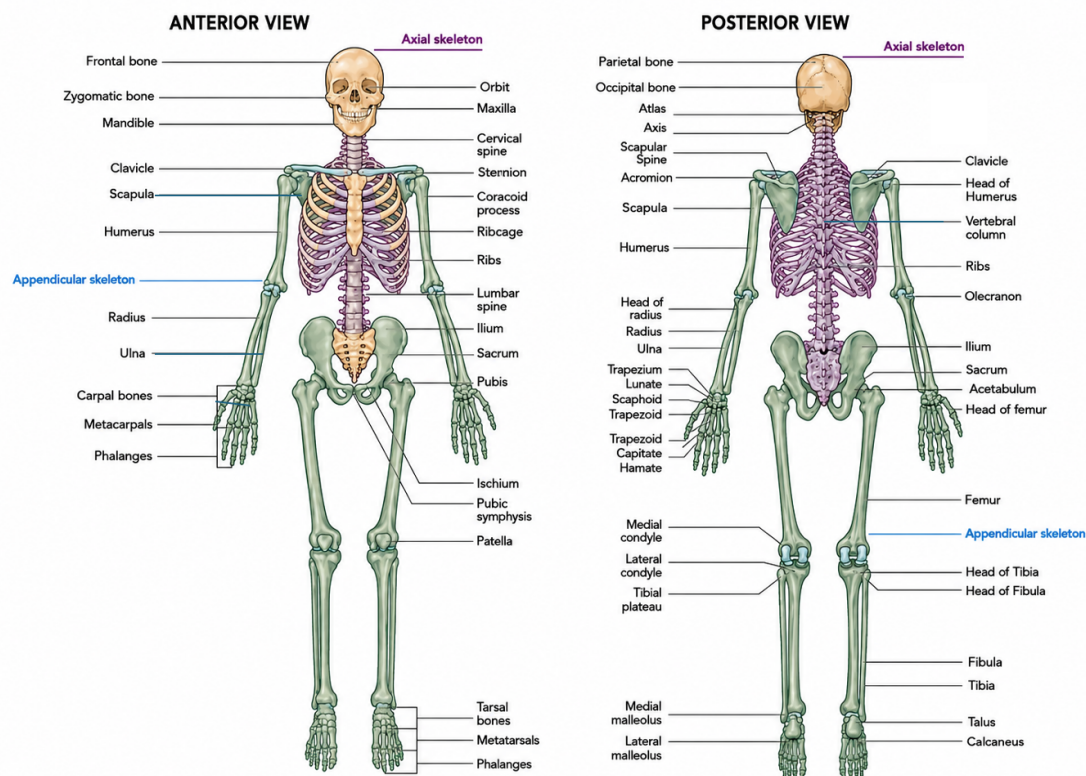
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COACHING INSIGHT

You don't need to memorise every anatomical term. What matters is understanding how the systems connect — so when a climber says their forearms 'pump out' on a route, you know what is happening physiologically and can adjust their training accordingly.

Section 3: The Skeletal System



Full labeled human skeleton, anterior and posterior views

The human skeleton is a collection of approximately 206 differently shaped bones that align to create a protective framework for the body that muscles can connect to and provide movement from. The skeleton serves four major functions:

1. **Support** — provides a framework to support the organs and tissues of the body
2. **Protection** — protects internal organs (skull protects the brain; thorax protects heart and lungs)
3. **Movement** — provides a framework for muscles to attach to; when muscles contract they pull on bones which act like levers to create movement
4. **Supply & Storage** — bones produce red and white blood cells in the bone marrow, and store minerals including calcium

3.1 Bone Structure

Bone is a rigid living organ with its own nerves and blood vessels. Its construction allows it to remain lightweight yet strong. Each bone is formed from multiple layers of three types of tissue:

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- **Compact bone** — heavy, dense, and strong; thickest at the bone's weakest point; provides primary structural strength
- **Cancellous bone** — honeycombed appearance; strong and hard without the weight of compact bone; absorbs compressive loads
- **Periosteum** — the membrane lining the outer surface of bones; houses nerves and blood vessels; attachment point for tendons and ligaments

Bone is constantly being broken down and rebuilt throughout life — bone remodelling — allowing the skeleton to adapt to mechanical loading, repair micro damage, and regulate blood calcium levels. This is maintained by four specialised cell types:

- **Osteogenic Cells** — stem-like precursors that divide to produce osteoblasts
- **Osteoblasts** — bone-building cells that initiate mineralisation
- **Osteocytes** — mature bone cells embedded in the matrix; maintain daily bone metabolism
- **Osteoclasts** — resorb old bone during remodelling and fracture repair

3.2a The Axial and Appendicular Skeleton

Division	Components & Climbing Relevance
Axial skeleton	Skull, vertebral column, and bony thorax. Protects the central nervous system and vital organs. In climbing, the vertebral column must maintain stiffness and stability, particularly on overhanging terrain.
Appendicular skeleton	Shoulder girdle, arms, hands, pelvic girdle, legs, feet. Allows movement through arm and leg joints. The shoulder girdle, upper limb, and finger bones are most critical for climbing; pelvis and lower limb drive footwork and heel hooks.

3.2b Types of Bone

Term	Definition / Description
Long bones	Cylindrical shaft with expanded ends acting as levers. E.g. femur, humerus, ulna.
Short bones	Small, cube-shaped, high surface area for shock absorption. E.g. carpals, tarsals.
Flat bones	Thin and often curved; protect organs, large muscle attachment area. E.g. ribs, ilium, scapula, skull.
Irregular bones	Complex shapes for multiple functions. E.g. vertebrae, pubis, maxilla.
Sesamoid bones	Embedded within tendons to protect joints and improve mechanical advantage. E.g. patella.

3.3 The Vertebral Column

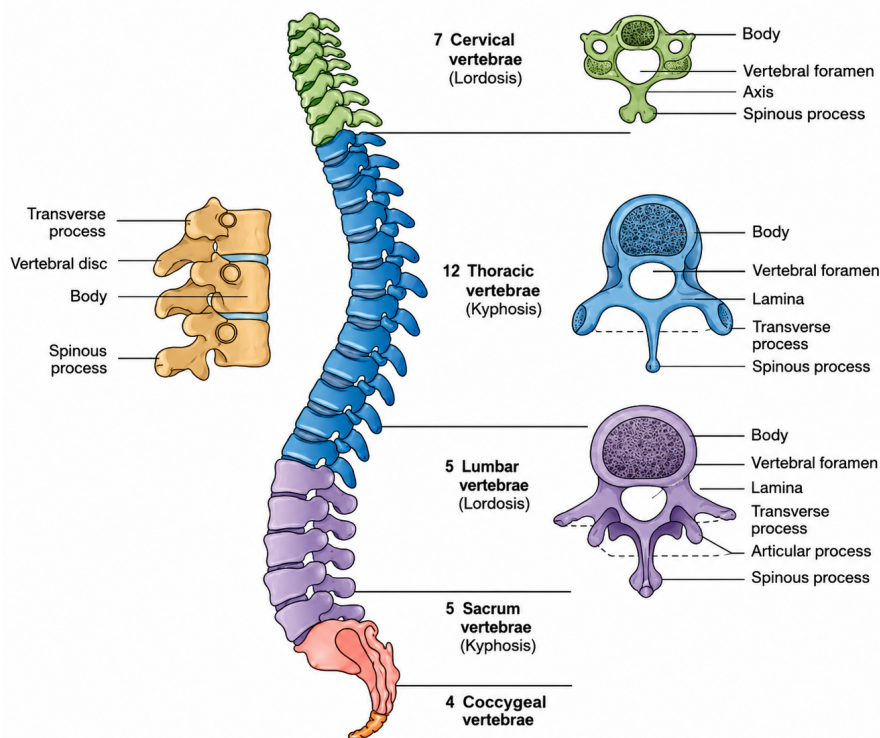
The vertebral column is a key part of the axial skeleton and a common site of injury and postural issues. It is made up of 33 individual vertebrae grouped into five sections:

Section	Number of Vertebrae & Coaching Notes
Cervical (C1–C7)	7 vertebrae — forms the neck. C1 and C2 allow head rotation. Climbers frequently develop stiffness here from looking up repeatedly.

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Section	Number of Vertebrae & Coaching Notes
Thoracic (T1–T12)	12 vertebrae — articulates with the rib cage. Thoracic extension and rotation are critical for overhangs and drop-knee technique. Loss of thoracic mobility is a common finding in climbers.
Lumbar (L1–L5)	5 vertebrae — the lower back. Must remain stable to transfer force between upper and lower body. Excessive flexion under load is a common fault and injury risk.
Sacrum (S1–S5)	5 fused vertebrae — connects the spine to the pelvis. Transmits ground reaction forces up through the kinetic chain.
Coccyx	4 fused vertebrae — the tailbone. Minimal functional role in climbing.



Cervical, thoracic, lumbar, sacrum and coccygeal vertebrae

3.4 Key Bones for Climbing Coaches

The following bones are of particular importance for understanding climbing movement and injury risk:

- **Phalanges (finger bones)** — proximal, middle, distal segments; flexor tendons run along the palm-side, held by annular pulleys. Most commonly injured structures in climbing.
- **Metacarpals** — hand bones connecting fingers to wrist. Important in grip and load transfer.
- **Carpals** — eight wrist bones allowing the wrist's wide range of motion.
- **Radius and Ulna** — the two forearm bones. The radius is the biceps brachii insertion point, central to elbow flexion during pulling.
- **Humerus** — upper arm bone; attachment site for the deltoid, biceps, and rotator cuff.
- **Scapula** — highly mobile bone, attachment for 17 muscles; stability critical for shoulder health.
- **Clavicle** — connects the scapula to the sternum, part of the shoulder girdle complex.
- **Femur** — the largest bone in the body; hip range of motion determines high-stepping and drop-knee ability.

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Section 4: Connective Tissue & Joints

4.1 Cartilage

Cartilage is an avascular connective tissue composed of collagen and elastic fibres within a gel-like matrix. It supports soft tissues, provides smooth articulating surfaces, and enables long-bone growth.

- **Hyaline Cartilage** — moderate collagen content. Found on articular surfaces of synovial joints and rib-sternum attachments.
- **Elastic Cartilage** — higher elastic fibre content; resilient and flexible. Found in the external ear and epiglottis.
- **Fibrocartilage** — very dense collagen fibres; tough, resists compression. Found in intervertebral discs and the pubic symphysis.

4.2 Fasciae

Fasciae are sheets of fibrous connective tissue that envelop muscles and organs, providing insulation and protection. They also form conduits for nerves, blood vessels, and lymphatics, and act as important sensory organs — richly innervated with proprioceptors and nociceptors.

- **Superficial Fascia** — separates skin from underlying muscles.
- **Deep Fascia** — lines the body wall and limbs, compartmentalising muscle groups.

4.3 Tendons & Ligaments

- **Tendons** — attach muscle to bone; transmit contraction force to the skeleton to produce movement (e.g. Achilles tendon).
- **Ligaments** — attach bone to bone; provide static joint stability and guide safe range of motion. Damage causes sprains and compromises stability.

4.4 Joints

A joint is the point where two or more bones meet, balancing movement and stability. Joints that prioritise stability sacrifice mobility, and vice versa. There are three main categories, classified by degree of movement:

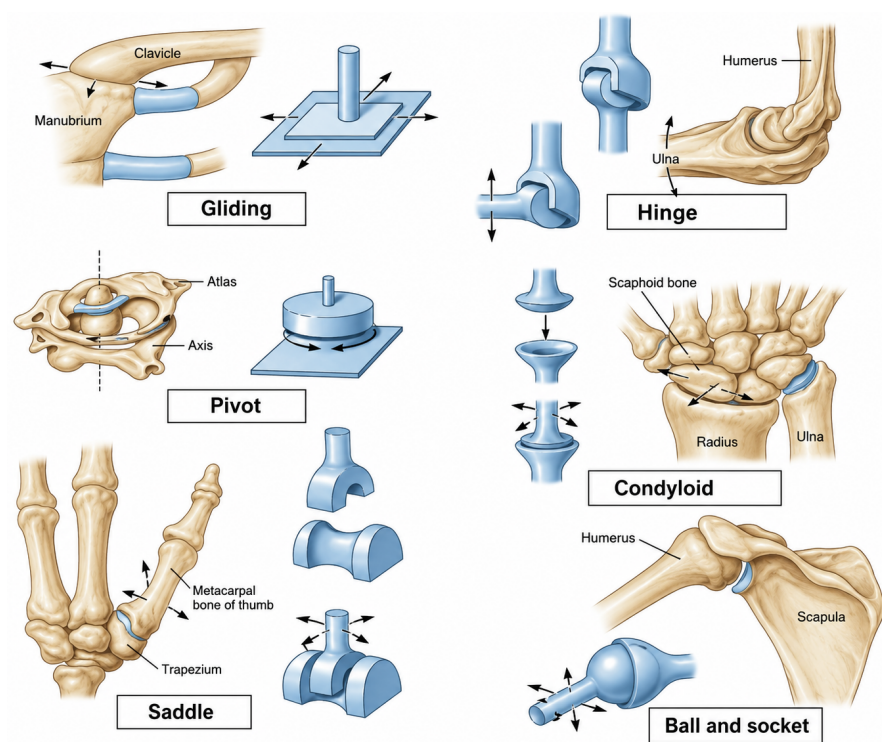
Joint Type	Structure	Movement & Climbing Example
Fibrous (Synarthrosis)	Bones joined by fibrous tissue	No movement — e.g. skull sutures.
Cartilaginous (Amphiarthrosis)	Bones joined by cartilage	Very limited movement — e.g. intervertebral discs, which absorb shock during landing and high-impact moves.
Synovial (Diarthrosis)	Fluid-filled joint capsule	Greatest range of movement — e.g. shoulder, elbow, hip, finger joints. The most important type for climbing coaches.

4.4b The Six Types of Synovial Joint

Synovial joints produce the greatest range of movement. There are six subtypes, all present in the climbing athlete's body:

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Gliding, hinge, pivot, condylod, saddle, and ball and socket joints

Synovial Joint Type	Movement Allowed	Climbing Example
Gliding (plane)	Sliding/gliding — side to side and back and forth	Between the carpal bones of the wrist; between the tarsal bones of the ankle
Hinge	Flexion and extension only	Elbow; interphalangeal finger joints; knee
Pivot	Rotation around a single axis	Atlas-axis joint (head rotation); radioulnar joint (pronation/supination)
Condylod	Flexion, extension, abduction, adduction, circumduction	Metacarpophalangeal joints (knuckles)
Saddle	Side to side and back and forth; no rotation	Carpometacarpal joint of the thumb — critical for pinch grip strength
Ball and socket	Multi-directional: flexion, extension, abduction, adduction, rotation, circumduction	Shoulder and hip — the two most mobile joints; critical for overhead reaching and high-stepping

4.4c Anatomy of a Synovial Joint

Understanding synovial joint structure helps coaches explain why injury occurs and what limits range of movement. The six key characteristics are:

1. **Articular cartilage** — smooth tissue covering joint surfaces; reduces friction, protects bone. Heals slowly due to poor blood supply.
2. **Articular capsule** — strong fibrous tissue surrounding the joint; adds stability, contains the joint cavity.

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3. **Synovial fluid** — lubricates surfaces, cushions the joint, provides nutrients for cartilage, absorbs friction debris.
4. **Ligaments** — connect bone to bone; control movement and stability. In fingers, collateral ligaments prevent sideways displacement.
5. **Articular discs (menisci)** — found in some joints (e.g. the knee); absorb shock and maintain stability.
6. **Bursae** — fluid-filled sacs (notably the shoulder) that reduce friction between tendons and bones. Bursitis is a common shoulder complaint in climbers.

4.4d Factors Limiting Synovial Joint Range of Movement

- **Joint articulation** — how bones fit together. The shoulder socket is shallow (mobile, needs muscular support); the hip socket is deep (stable, limited range).
- **Joint ligaments and capsule** — restrict unwanted movement. Tight ligaments reduce flexibility; overstretched ones reduce stability and raise injury risk.
- **Muscles and tendons around the joint** — provide dynamic stabilisation. The rotator cuff is the primary stabiliser of the shallow shoulder joint; weakness here raises injury risk significantly.
- **Temperature** — warm tissues are more extensible; a proper warm-up increases ROM.
- **Injury/Inflammation** — swelling, scar tissue, and pain guarding all reduce ROM around an injured joint.
- **Training history** — regular flexibility training increases ROM through neural adaptation and structural change.

KEY CLIMBING JOINTS

The finger PIP and DIP joints (hinge), the shoulder glenohumeral joint (ball and socket), the elbow (hinge), and the hip joint (ball and socket) are the four highest-priority joints for climbing coaches to understand. Most climbing injuries involve one of these four joints or the connective tissue immediately surrounding them.

End of Lesson 01. Return to corelearning.co.za to complete the Lesson 01 quiz (80% pass mark) and unlock the next lesson.